Operational Research

Introduction to Simulation Tutorial

The following exercises will require the use of Excel. An example question and excel spreadsheet is given on Course Resources.

Exercises

1) By Hand Complete the following table

Customer Arrival Time	Service Starts	Service Duration	Service Ends	Customer Waiting Time	Service Waiting Time
4		3			
6		3			
12		4			
14		2			
17		3			

2) Data has been collected at a service centre to enable the analysis and redesign of the service system.

Customers arrivals are described by the distribution

Inter-arrival Times	5	10	15	20	25	30
Frequency percentage	10	20	40	20	7	3

Service times are given by

Service Duration	10	12	14	16	20	30
Frequency percentage	5	20	45	20	9	1

Simulate the arrival of 30 customers, assuming that only the quoted times could occur, what does your simulation indicate about this system? What could be done to improve the system? *

Customer	**Inter-arrival Time (IAT)**	**Arrival Time (AT)**	**Service Starts (SS)**	**Service Duration (SD)**	**Service Ends (SE)**	**Customer Waiting Time**	**Service Waiting Time**
1	Generate using random number	IAT(1)	AT(1)	Generate using random number	SS(1) + SD(1)	SS(1) – AT(1)	SS(1)
2	Generate using random number	AT(1) + IAT(2)	Max{SE(1), AT(2)}	Generate using random number	SS(2) + SD(2)	SS(2) – AT(2)	SS(2) - SE(1)

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3) Jobs arrive every 15 minutes at a workshop comprising a single production stage.

Job arrivals are described by the distribution

Number of jobs arriving	5	7	11	12
Frequency percentage	10	20	40	30

Service capacities (i.e. the number of jobs processed per 15 minutes) are given by

Jobs Capacity (Service Capacity)	6	8	10	15
Frequency percentage	15	40	35	10

Simulate the production process over a period of 4 hours, what does your simulation indicate about this system? What could be done to improve the system? **

Time	**No. Jobs Arriving (JA)**	**Jobs in Queue (JQ)**	**Jobs Capacity (JK)**	**Jobs Completed (JC)**	**Total Jobs Completed (TJC)**	**Total Jobs Left (TJL)**	**Unused Capacity (UC)**
1 (0-15)	Generate using random number	JA(1)	Generate using random number	Min{JQ(1),JK(1)}	JC(1)	JQ(1)-JC(1)	JK(1)-JC(1)
2 (15-30)	Generate using random number	JA(2)+ TJL(1)	Generate using random number	Min{JQ(2),JK(2)}	JC(1)+JC(2)	JQ(2)-JC(2)	JK(2)-JC(2)

4. Try and write a simulation in Python to simulate the following:

a. Throwing a n-sided dice. Start with a 6-sided dice and then abstract. Your simulation should be able to compute the long-term behaviour of k trials, e.g. what is the observed probability of throwing a 1?
b. Try to see if you can simulate the above questions. I'll show some code next week to do this. You will want to use the `random` module in Python.

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